

Run B Record

RUN B — THERMAL RESISTANCE CONTROL

CIP-BTSB-BT-001 · Supporting Record R-B

CIP — Compute Infrastructure Platform · S. Mohammed Idreesh · Chennai Run date: 27 August 2026 · Issued before Run A was performed

[PRE-GEN-0 MECHANISM DEMONSTRATION — NOT A PRODUCT VALIDATION]

1. PURPOSE

Run B is the **control**. It places a 3 mm liquid layer in the heat path that is chemically similar to paraffin wax but **does not change phase** in the temperature range of the test.

Its function is to separate two effects that both produce delay:

C → B	delay caused by a LAYER BEING IN THE WAY	(thermal resistance)
B → A	delay caused by MELTING	(latent heat)

Without Run B, any delay measured in Run A could not be attributed to phase change. **Run B is what makes Run A interpretable.**
Issued before Run A so that the resistance baseline is fixed and published while the latent-heat measurement does not yet exist.

2. CONFIGURATION AS RUN

Item	Detail
Layer	Liquid paraffin (mineral oil), 25.6 mL , syringe-metered
Cavity	155 × 55 × 3 mm silicone gasket, aluminium foil lined — §3
Plates	H 98.24 g, C 97.83 g · pair 172.5 J/K
Heaters	3 × ceramic cartridge in series, 13.7 W electrical
Clamps	4 × 2" G-clamp, 43 / 60 / 105 / 130 mm, finger tight + ½ turn
Insulation	Full ceramic-fibre enclosure, clamps wrapped — as Run C-2
Orientation	Heated from above

3. APPARATUS CHANGE — ALUMINIUM FOIL LINER

Recorded in full, including the failure that caused it.

3.1 What happened

The first attempt to fill the cavity leaked. Oil escaped continuously past the gasket at room temperature. Diagnosis: oil had contaminated the gasket’s sealing faces during pouring, destroying the silicone-to-aluminium grip, and a completely full cavity left the displaced oil nowhere to go but out.

Degreasing and re-clamping were attempted. The leak persisted.

3.2 The fix

An 11 µm aluminium foil liner (single layer, ISI IS 15392:2003, grade 40800) was pressed into the gasket window and folded over the gasket rim, forming a continuous foil pocket. The oil sits inside the foil. A five-minute static leak test after clamping showed no escape.

Alternatives considered and rejected:

Option	Rejected because
Kapton / PVC tape on the outside	Adhesive does not bond to oil; PVC softens at 80 °C; external tape cannot restrain internal pressure
RTV silicone sealant	24 h cure; bonds the plates together, requiring the mating faces to be scraped before Run A; and a hand-laid bead changes gasket thickness by an unrepeatable amount, making the cavity volume differ between Runs B and A

The last point governed the decision. **Cavity depth is a controlled variable (§5.3) and sets the wax charge for Run A.** Foil adds 0.011 mm — 0.4% of the gasket — identically in both runs.

3.3 Consequence — quantified in §5

The foil solves the leak but introduces a **thermal bypass**, which was identified during the run and is quantified in Section 5. **The same liner is used in Run A**, so the bypass is present in both runs being compared.

4. RESULTS

4.1 Timing

Power ON	11:40 clock · t = 70 s
TC2 = 85.0 °C	12:17 clock · t = 2315.5 s
Samples	2,501 · zero sensor faults
Peak TC1	92.25 °C — never approached the 110 °C early-stop
Peak TC2	86.25 °C

4.2 Baseline

TC1 37.96 °C	TC2 38.26 °C
TC1 – TC2 = –0.302 °C	sd 0.316

Ambient (mean plate) **38.11 °C**, against Run C’s **36.19 °C** — a difference of **1.92 °C**. §5.3 requires runs to start within ±2 °C of one another. **This run satisfies that condition, but only just, and it is recorded as marginal.**

4.3 PRIMARY ENDPOINT

t85 (Run B) = 2245.5 s = 37.42 minutes
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By linear interpolation between the straddling samples, per §6.1.

4.4 RESISTANCE DELAY — the result this run exists to produce

t85 (Run C)	=	22.04 min
t85 (Run B)	=	37.42 min
Δt_resistance	=	923 s = 15.38 minutes

This is the delay produced by a 3 mm non-melting layer. Any delay Run A shows beyond 37.42 minutes is attributable to latent heat.

4.5 Steady-state temperature drop across the layer

TC1 – TC2 = 7.38 °C	sd 0.34
(measured from 600 s after power-on to t85)	

Both plates rose at an identical rate through this interval with a constant separation — the signature of steady-state conduction through a fixed thermal resistance.

For comparison, Run C’s direct-contact figure was **2.23 °C**.

4.6 Heat model

	RUN C	RUN B
P into plates	9.49 W	7.26 W
Loss coefficient	0.1061 W/K	0.1114 W/K
Time constant	27.1 min	25.8 min
Ceiling (T_mean)	126 °C	103 °C

Loss coefficients agree to within 5%, confirming the insulation was rebuilt consistently between runs.

The operator observed during the run that the G-clamps were warm to the touch. That observation is the qualitative form of the 0.111 W/K figure: the clamps are steel in direct contact with a heated plate, projecting into ambient air, and they are a principal loss path. It is recorded because it was noticed independently of the fit.

5. THE FOIL BYPASS — QUANTIFIED LIMITATION

Identified during the run, from the observation that TC1 – TC2 stabilised at 8 °C instead of climbing toward the predicted 22-28 °C.

The foil folds over the gasket rim, descends the 3 mm inner wall, and reaches the opposite plate. This forms an **unbroken aluminium path around the entire window perimeter, in parallel with the oil layer.**

Window perimeter	0.420 m
Foil thickness	11 µm
Bypass cross-section	4.62 × 10 ⁻⁶ m²
Path length	3 mm
k (aluminium)	200 W/m·K
OIL layer conductance	0.398 W/K
FOIL bypass conductance	0.308 W/K
Total	0.706 W/K
At ΔT = 7.38 °C:	5.21 W crosses the gap
	2.94 W through the oil
	2.27 W through the foil
THE FOIL CARRIES 44% OF THE HEAT CROSSING THE LAYER.	

5.1 Effect on the results

The bypass makes heat cross faster than the oil alone would allow. Therefore:

- **Δt_resistance is UNDERSTATED.** The true resistance delay of 3 mm of oil is larger than 923 s.
- **Δt_latent (A – B) will also be understated,** since the same bypass is present in Run A.

Both errors bias the results downward — against the mechanism, not in favour of it. The figures reported here are conservative.

5.2 Why the run was not repeated

The bypass was identified at approximately 10 minutes into the run. Aborting would have required a new sealing method, and the alternatives in §3.2 were rejected for reasons that had not changed. **The configuration is fully characterised, identical in Runs B and A, and biased conservatively.** It is reported rather than concealed.

6. RESULTS AGAINST PREDICTIONS

Quantity	Predicted §7.2	Measured	Verdict
Δt_resistance	100 – 250 s	923 s	✗ 3.7× above the band
TC1 – TC2 steady, Run B	22 – 28 °C	7.38 °C	✗ a quarter of the band
t ₈₅ Run B	13 – 19 min	37.42 min	△ longer, as §7.2 anticipated
Run B terminates early on TC1	predicted likely §7.3	did not occur — peak TC1 92.25 °C	✗ prediction wrong

Three predictions failed and one behaved as anticipated. All are reported.

7. INTERPRETATION

At first reading, two of the failures appear contradictory: **the layer delayed heat far more than predicted, while the temperature drop across it was far smaller than predicted.** Both cannot follow from the §7 model, which assumed roughly 15 W crossing a 3 mm oil layer.

Two measured facts resolve it.

The small ΔT follows from lower power and the foil bypass. Run C measured delivered power at **7.73 W**, not 15 W, and the foil carries 44% of what crosses. Less power through the layer, smaller temperature drop.

The long delay is not principally the layer’s own resistance. It is that the layer **forces the hot plate to run hotter for the entire run**, and a hotter plate loses more heat to ambient. The cold plate is starved. In Run C the plates sat 2.23 °C apart; in Run B, 7.38 °C apart for thirty-seven minutes, with the hot side shedding heat throughout.

The pre-registered model treated the layer as a passive resistance in a closed system. The measurement shows it also acts as a loss amplifier in an open one. That is a finding about the fixture, and it is more informative than a correct prediction would have been.

8. IMPLICATION FOR RUN A

Run A must complete within the **45-minute limit** established in Record R-C.

Run B non-plateau path	37.4 min
Wax conducts better than oil (0.25 vs 0.14)	→ shorter
Plus the melting plateau	→ 13 - 25 min
Estimated t85 (Run A)	43 - 55 min

Run A may not reach 85 °C inside the limit.

The limit will not be extended a second time. It was set at 45 minutes on 26 August, published with its reason, before Runs B and A existed. Moving it again after seeing Run B’s timing would be the precise failure that pre-registration exists to prevent.

If t85(A) is not reached, the result is reported as a lower bound:

t85(A) > 45 min > 37.42 min	→	Δt_latent > 456 s
H1 is falsified if Δt_latent ≤ 60 s.		
A lower bound of 456 s exceeds that threshold by a factor of 7.6.		

H1 would be confirmed by a bounded result. Separately, the plateau — the primary evidence for the phase-change mechanism, expected at 58–68 °C — occurs well within the run window and will be observed regardless of whether the endpoint is reached.

9. STATUS

[✓]	RUN C	direct contact	t85 = 22.04 min
[✓]	RUN B	liquid paraffin	t85 = 37.42 min
[]	RUN A	paraffin wax 23.0 g	
Δt_resistance = 923 s MEASURED			
Δt_latent = pending Run A			

Raw data published as `run_B_20260827.csv`, unedited.

Record R-B · Issued 27 August 2026, before Run A. Supports CIP-BTSB-BT-001. All build-manual performance figures remain [GEN-0 VALIDATION REQUIRED].